CSCI 693 Paper Template*

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Abstract

The Abstract should offer a succinct summary of the entire paper, usually 150–300 words. Key contents: Research problem, methods, main findings, and major conclusions.

1 Citation

Appropriately cite all references to other published works included in the paper. `biblatex` is used to create a list of references or bibliography as the last section in the paper. Here are citation examples for a book[2], a journal paper[1], a website[3], and a conference proceeding paper[4]. Please check out the source code of this document for details.

2 Style

Write clearly and simply in the third person for an audience that is well-grounded in computing, but who may have limited exposure or knowledge about the specific topic of your paper. Define any technical terms deemed to require clarification when they are introduced.

3 Introduction

Use sections and subsections to guide the reader. If the section title shown contains an "or", you may choose to use either name (do not use both). You may adapt the subsections as needed.

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3.1 Context and Background

Present general information on the topic. Highlight the broader context and relevance of the research.

3.2 Problem Statement and Research Question

Clearly state the gap in knowledge or the problem your study addresses. Articulate the main research question(s) or hypothesis/hypotheses.

3.3 Significance of the Study

Explain why your research is necessary or valuable. Discuss theoretical, practical, or policy implications.

3.4 Scope and Organization

Briefly outline what the reader can expect in the upcoming sections.

4 Literature Review

Sometimes it may be helpful to use lists. Lists are easy to create in LaTeX whether they are ordered, unordered, or nested as shown in the following example.

- Overview of Existing Research
- Critical Evaluation

1. Theoretical Framework (If Applicable)
2. Research Gap and Rationale

1. Overview of Existing Research
    - Summarize and synthesize relevant scholarly work.
    - Identify trends, debates, and major theories.
2. Critical Evaluation
    - Critically analyze strengths and weaknesses in the current body of literature.
    - Highlight methodological gaps or inconsistencies.
3. Theoretical Framework (If Applicable)
    - Present the theories or conceptual models guiding your research.
    - Show how your study aligns with or challenges existing theories.
4. Research Gap Rationale
    - Pinpoint the specific knowledge gap your research aims to fill.

5 Methodology (or Materials and Methods)

5.1 Research Design

Describe whether the study is experimental, correlational, qualitative, mixed methods, etc.

Participants and Setting: Detail demographic information (if relevant). Specify the sampling method (probability, purposive, convenience, etc.).

Data Collection Methods: Explain instruments (e.g., surveys, interview protocols, laboratory equipment) and procedures. Justify your choices and address reliability and validity.

Data Analysis Techniques: Describe statistical tests, qualitative coding strategies, or computational models used. Justify why these analyses are appropriate for your data.

5.2 Ethical Considerations

Note institutional review board (IRB) approval if required. Describe how you ensured confidentiality, consent, and data protection.

5.3 Code Listings

Short, relevant code snippets or pseudocode can be included inline if they are important for enhancing understanding, provided they are well-formatted and clearly labeled. Commands from `listings` package allow you to display code easily with customizable coloring and styling rules. Here is an example.

Listing 1: Python example

```
x = 42
epsilon = 0.01
step = epsilon**2
num_guesses = 0
ans = 0.0
while abs(ans**2-x) > epsilon and ans < x:
    ans = ans + step
    num_guesses += 1
if abs(ans**2-x) <= epsilon:
    print(str(ans) +
    ' is close to the square root of ' +
    str(x))
else:
    print('Failed to find square root of ' + str(x))
print("The number of guesses is " + str(num_guesses))
```

6 Results (or Findings)

6.1 Presentation of Data

Organize findings logically (chronologically, thematically, or by research questions). Use tables, figures, charts, or direct quotes (for qualitative research) as needed. Table 1 is an example of a simple table.

Table 1: Nonlinear Model Results

Case	Method#1	Method#2	Method#3
1	50	837	970
2	47	877	230
3	31	25	415
4	35	144	2356
5	45	300	556

6.2 Figures

The "`figure`" environment should be used for figures. One or more images can be placed within a figure. If your figure contains third-party material, you must clearly identify it as such, as shown in the example Figure 1 below.

Your figures should contain a caption which describes the figure to the reader. Figure captions are placed *below* the figure.

Every figure should also have a figure description unless it is purely decorative. These descriptions convey what's in the image to someone who cannot see it. They are also used by search engine crawlers for indexing images, and when images cannot be loaded.

A figure description must be unformatted plain text less than 2000 characters long (including spaces). **Figure descriptions should not repeat the figure caption – their purpose is to capture important information that is not already provided in the caption or the main text of the paper.** For figures that convey important and complex new information, a short text description may not be adequate. More complex alternative descriptions can be placed in an appendix and referenced in a short figure description. For example, provide a data table capturing the information in a bar chart, or a structured list representing a graph. For additional information regarding how best to write figure descriptions and why doing this is so important, please see `https://www.acm.org/publications/taps/describing-figures/`.

Figure 1: A screenshot of the music video for "Never gonna give you up", performed by Rick Astley. Screenshot via Wikipedia. (`https://en.wikipedia.org/wiki/Rickrolling`).

6.3 Key Observations

Summarize significant findings without interpreting them in depth—save interpretation for the Discussion section (unless Results and Discussion are combined).

6.4 Statistical or Thematic Findings

Report p-values, effect sizes, confidence intervals, or themes, depending on methodology.

7 Discussion

7.1 Interpretation of Results

Connect your findings to your original research questions. Explain how the results support or challenge existing literature/theories.

7.2 Implications

Discuss practical, theoretical, or policy implications. Show how your work advances knowledge in the field.

7.3 Limitations

Acknowledge constraints such as sample size, method-specific limitations, or biases. Suggest how these limitations affect interpretation and generalizability.

7.4 Suggestions for Future Research

Propose follow-up studies or alternative methods that address unanswered questions.

8 Conclusion

- Summary of Main Findings: Recap key points succinctly.
- Research Contribution: Reiterate how the study fills the identified gap.
- Closing Thoughts: Provide a final reflection on the significance and potential impact of your research.

We're no strangers to love. You know the rules and so do I. A full commitment's what I'm thinking of. You wouldn't get this from any other guy. I just wanna tell you how I'm feeling. Gotta make you understand.

Never gonna give you up. Never gonna let you down. Never gonna run around and desert you. Never gonna make you cry. Never gonna say goodbye. Never gonna tell a lie and hurt you.

We've known each other for so long. Your heart's been aching, but you're too shy to say it. Inside, we both know what's been going on. We know the game and we're gonna play it. And if you ask me how I'm feeling. Don't tell me you're too blind to see.

Never gonna give you up. Never gonna let you down. Never gonna run around and desert you. Never gonna make you cry. Never gonna say goodbye. Never gonna tell a lie and hurt you.

References

[1] Albert Einstein. "Zur Elektrodynamik bewegter Körper. (German) [On the electrodynamics of moving bodies]". In: *Annalen der Physik* 322.10 (1905), pp. 891–921. DOI: `http://dx.doi.org/10.1002/andp.19053221004`.

[2] Michel Goossens, Frank Mittelbach, and Alexander Samarin. *The LaTeX Companion*. Reading, Massachusetts: Addison-Wesley, 1993.

[3] Donald Knuth. *Knuth: Computers and Typesetting*. `http://www-cs-faculty.stanford.edu/~uno/abcde.html`.

[4] Frank Maurer. "Agile Methods and Interaction Design: Friend or Foe?" In: *Proceedings of the 1st ACM SIGCHI Symposium on Engineering Interactive Computing Systems*. EICS '09. Pittsburgh, PA, USA: ACM, 2009, pp. 209–210. ISBN: 978-1-60558-600-7. DOI: `10.1145/1570433.1570435`. URL: `http://doi.acm.org/10.1145/1570433.1570435`.